Data Analytics for Sustainability (A25)

ELECTRIC VEHICLE POPULATION DATA ANALYSIS

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1. Introduction

This report presents data-analysis of the Electric Vehicle (EV) Population dataset using Python-based statistical and machine learning techniques. It investigates relationships between car features like Model Year, Base MSRP, Manufacturer (Make), and Electric Range to understand the EV market, technological progress, and pricing trends. The workflow includes data preparation, exploratory data analysis (EDA), visualization, regression modelling, and clustering.

2. Data Cleaning and Preparation

The raw dataset was assessed for duplicate rows, missing values, improper data types, and outliers. Base MSRP, Model Year, and Electric Range were ensured as numeric and Make, Model, County as categorical. Zero values in Base MSRP were replaced by the mean MSRP grouped by Electric Vehicle Type. This was supported by the assumption based on data collected from reliable published journals. Upon further analysis, no duplicates rows were found. Thus, the dataset was cleaned for analysis, ensuring accuracy and reliability for further modelling.

3. Exploratory Data Analysis (EDA)

In the dataset, there are two types of EVs: Plug-in Hybrid Electric Vehicle (PHEV) and Battery Electric Vehicle (BEV). BEVs form 76.2% and PHEVs form 23.8% of the dataset. Therefore, fully electric models dominate the market. The mean Base MSRP for PHEV is USD 53148.88 and for BEV is USD 59734.03 with a standard deviation of USD 4885.57. The lowest price of an EV is USD 31950, and the highest price is USD 845000. Electric Range varies from 6 to 330 miles with two clusters: short-range PHEVs and long-range BEVs with Model Year spanning from 1997 to 2022. Top Manufacturers are Tesla, Nissan, and Chevrolet. King County (41.5%), Snohomish (17.2%), and Pierce (11.8%) had the highest registrations, suggesting urban infrastructure such as charging stations and possibly, higher income compared to rural regions.

Table 1.

Linear Correlation between Model Year, Base MSRP and Electric Range of EVs

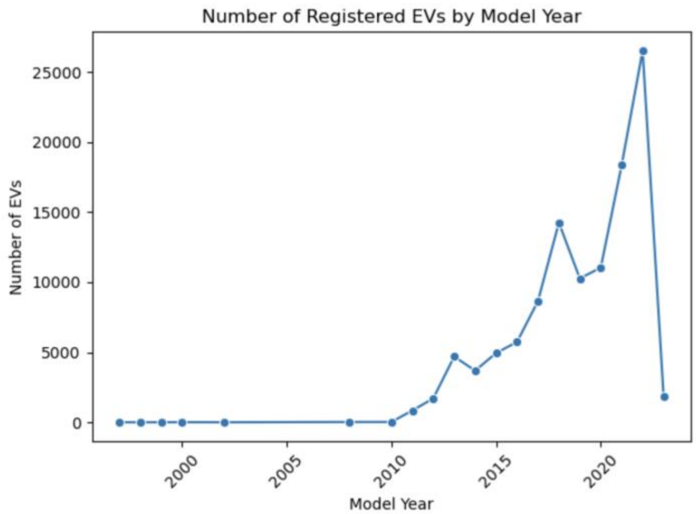
Relationship	r	Interpretation
Model Year and Base MSRP	0.077	Very weak positive, newer vehicles slightly more expensive.
Model Year and Electric Range	–0.288	Weak negative, newer years include many short-range PHEVs.
Base MSRP and Electric Range	0.213	Weak positive, higher-priced vehicles have marginally higher ranges.

No strong linear relationships exist, supporting the need for multivariate modelling to analyse nonlinear relationships.

4. Data Visualization

Figure 1.

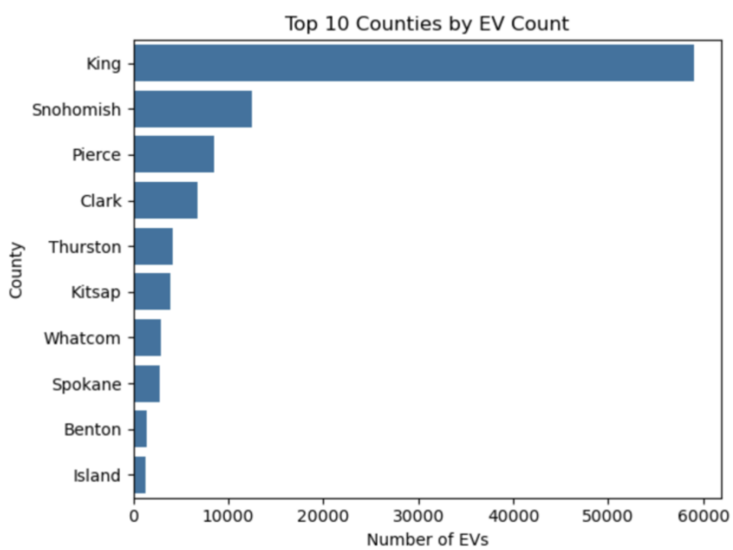
Number of Registered EVs by Model Year



This plot visualizes how the number of electric vehicle registrations has changed across model years, effectively showing the growth trajectory of EV adoption over time. A line chart is ideal for temporal data because it highlights trends and inflection points over continuous years. By connecting data points with lines, we can easily observe patterns of increase or stagnation in vehicle registrations. There is exponential growth after 2015, reflecting the commercialization of affordable EVs (e.g., Tesla Model 3, Nissan Leaf). The sharp peak in 2021–2022 demonstrates heightened consumer demand and supportive policy environments. Before 2010, EV registrations were minimal, confirming the nascent stage of EV technology.

Figure 2.

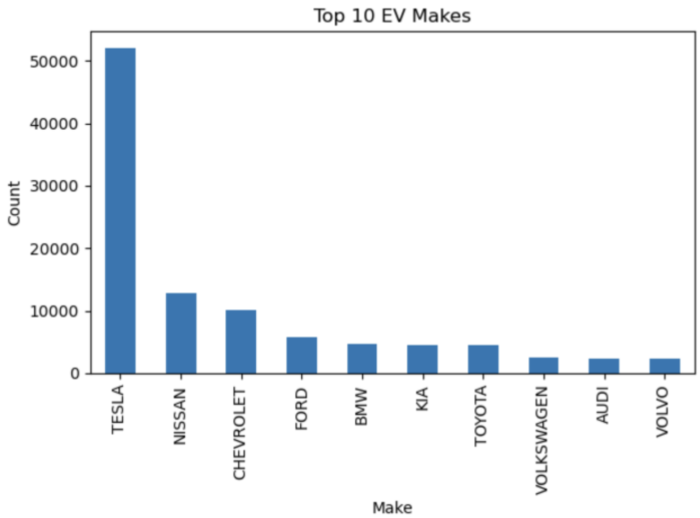
Top 10 Countries by EV Count



This plot illustrates the geographical distribution of electric vehicle registrations and identify regions with the highest concentration of EVs. A horizontal bar chart provides a clear visual ranking of categorical data (counties) by magnitude (number of EVs). It emphasizes comparative differences between areas and highlights regions leading in EV adoption. King County has the highest EV count, followed by Snohomish and Pierce counties. This indicates that urbanized, high-income regions tend to adopt EVs earlier, likely due to better infrastructure and environmental awareness. Rural counties show lower adoption, possibly due to limited charging networks and higher upfront costs.

Figure 3.

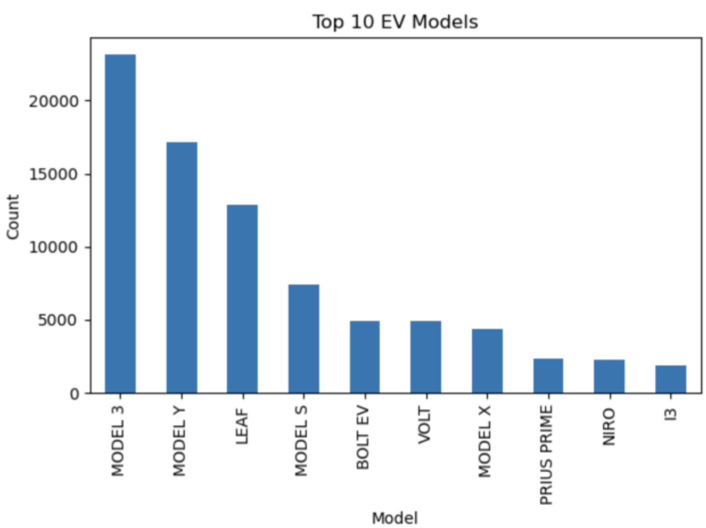
Top 10 EV Makes



This bar graph shows which automobile manufacturers dominate the EV market in the dataset. A vertical bar chart effectively communicates market share distribution among manufacturers. It allows a quick visual comparison of production volume or registration frequency. Tesla leads by a significant margin, followed by Nissan and Chevrolet. This reflects Tesla's technological leadership and brand influence in the EV industry. The presence of multiple other brands (Ford, BMW, Kia, Hyundai) highlights market diversification and increasing competition.

Figure 4.

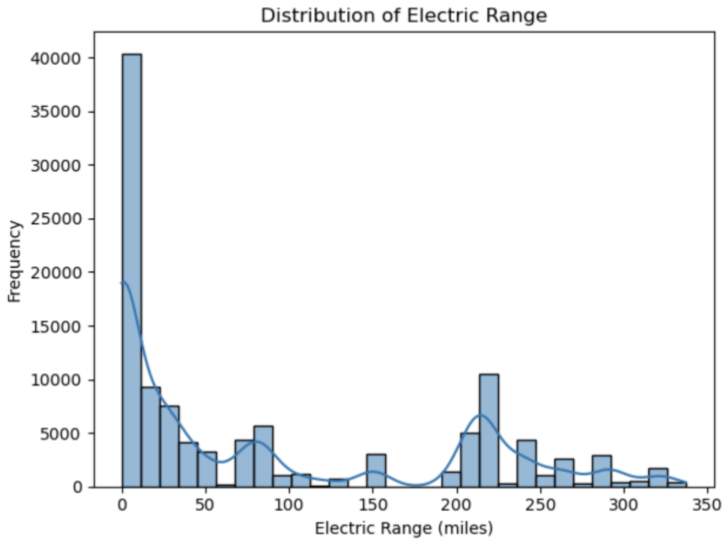
Top 10 EV Models



This bar graph identifies which specific EV models are most popular or widely adopted. By displaying model-level data, this visualization helps differentiate consumer preferences and product success across vehicle categories. A bar chart succinctly communicates ranking and relative frequency. The Tesla Model 3 and Nissan Leaf emerge as the most popular models. These vehicles represent different price tiers : Tesla (premium, high-range) and Nissan (affordable, moderate range). It demonstrates the coexistence of both mass-market and luxury EV segments in the growing market.

Figure 5.

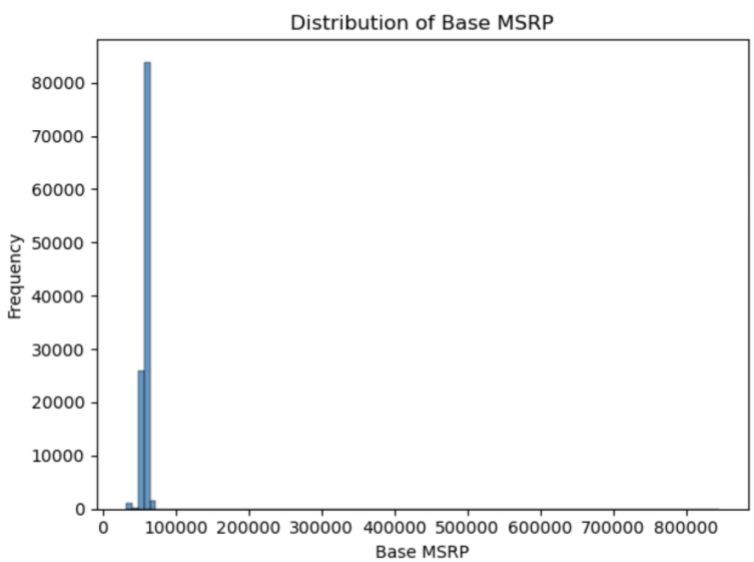
Distribution of Electric Range



This histogram visualizes how electric range values (distance per charge) are distributed across the dataset. A histogram with a Kernel Density Estimate (KDE) overlay shows both frequency and shape of distribution. It's the best method for understanding the spread, central tendency, and skewness of continuous numeric data. The distribution is bimodal: one peak between 20–50 miles (PHEVs) and another between 200–300 miles (BEVs). Confirms technological differences between vehicle types. Indicates overall improvement in range performance over time due to advancements in battery technology.

Figure 6.

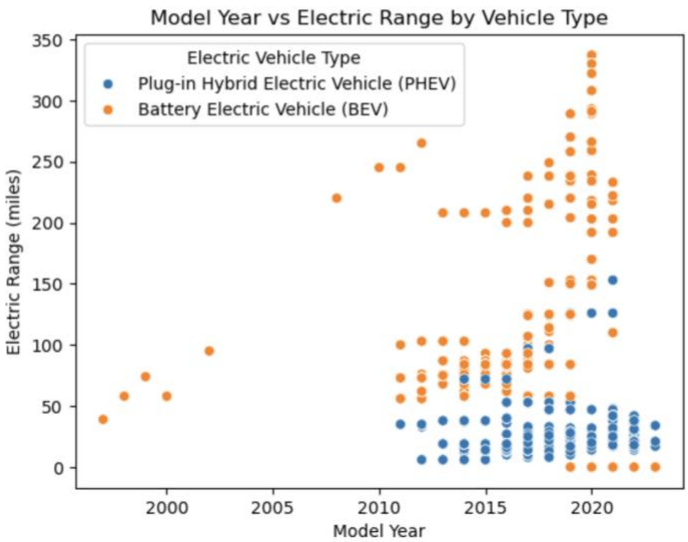
Distribution of Base MSRP



This histogram analyzes the price distribution of electric vehicles. A histogram reveals price concentration and market segmentation (affordable vs. premium EVs). It also helps detect skewness and outliers that could distort average values in analysis. Prices are right-skewed: most EVs are priced between USD 30,000–USD 70,000, while a few luxury models exceed $100,000. This confirms that EV markets are expanding beyond premium buyers to middle-income consumers and supports later regression findings that higher price correlates with greater electric range.

Figure 7.

Model Year vs Electric Range by Vehicle Type



This scatterplot examines the relationship between a vehicle's production year and its electric range, and to assess how technology has evolved over time. A scatter plot is ideal for exploring bivariate relationships between two continuous variables. The addition of a color hue (vehicle type) introduces a categorical dimension, revealing patterns within subgroups. A clear upward trend shows that newer EV models achieve longer ranges. BEVs (colored separately) dominate the upper portion of the plot, confirming their superior performance compared to PHEVs. The plot ongoing technological progress in battery capacity and energy efficiency and validates the positive relationship later quantified in the regression model between Model Year and Electric Range.

5. Regression Analysis

To understand the effect of Model Year, Base MSRP and Make on Electric Range, regression analysis was performed. The three attributes were encoded, and dataset was split into 80% for training and 20% for testing.

Model Performance:

- $R^2 = 0.774$. This explains 77% of variance in Electric Range is attributed to these three features.
- RMSE = 48.64 miles, i.e. an error of 48.64 miles which is a strong predictive accuracy
- The beta values indicate how much Electric Range changes when one predictor increases by one unit, keeping other variables constant. These were derived using linear algebra (OLS estimation) performed by scikit-learn's LinearRegression() function.

1. Coefficients:

Table 2.

β values for Model Year, Base MSRP in USD and Make of the EV

Variable	β	Interpretation
Model Year	+4.87	Each additional year increases the expected Electric Range by 4.87 miles, keeping MSRP and Make constant.
Base MSRP ($)	+0.0035	Each additional $1 in MSRP increases range by 0.0035 miles, or roughly +3.5 miles per $1,000.
Make	Varies	Dummy variables for each manufacturer were allocated. Tesla's dummy had the highest β, meaning Tesla vehicles tend to have longer ranges than others at similar price/year.

6. Clustering Analysis

The clustering results reveal clear distinctions among electric vehicle segments when plotted by Electric Range and Base MSRP. Each cluster represents a unique group of vehicles sharing similar performance and pricing characteristics.

Table 3.

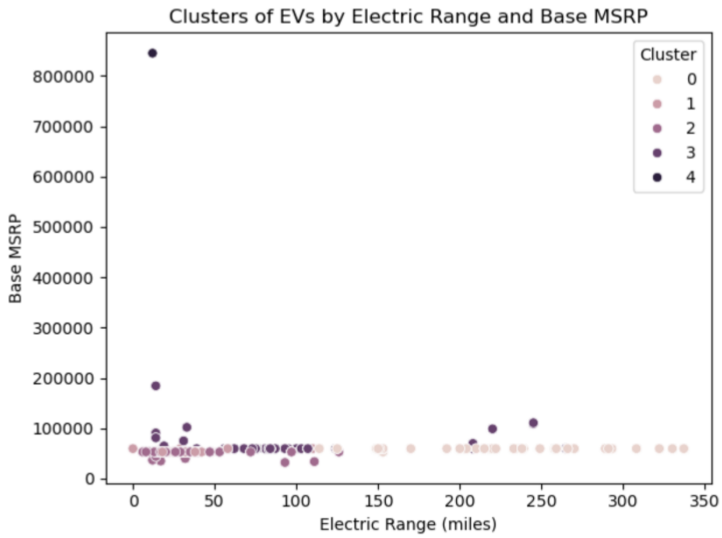
Clustering Analysis based on Mean Electric Range (miles), Mean Base MSRP and Typical Model Year.

Cluster	Mean Electric Range (miles)	Mean Base MSRP ($)	Typical Model Year	Interpretation
Cluster 0	231.36	59,733	2018	Long-range BEVs (e.g., Tesla, Nissan, Chevrolet): high performance and mainstream appeal.
Cluster 1	4.78	58,725	2022	Recent plug-in hybrids with very short electric ranges: rely mainly on gasoline backup.
Cluster 2	34.05	52,148	2017	Transitional PHEVs with moderate price and range: bridge technology between hybrids and BEVs.
Cluster 3	107.91	61,350	2015	Mid-generation EVs with moderate range and cost: represent earlier BEV models.
Cluster 4	12.00	845,000	2015	Outlier cluster containing an extremely high-priced car or a data anomaly.

This segmentation confirms a strong positive relationship between vehicle range and price.

Figure 8.

Cluster of EVs by Electric Range and Base MSRP



The scatter plot of Electric Range versus Base MSRP colored by K-Means cluster assignment visually depicts the segmentation of electric vehicles across range and price dimensions. Distinct groups are observed, corresponding to high-range BEVs, mid-range transitional models, and low-range hybrids. The plot reveals a positive relationship between vehicle range and price, consistent with the cost of higher-capacity battery technology. A single extreme outlier, isolated at a price exceeding $800,000 and a minimal range, demonstrates the algorithm's ability to detect anomalous entries. Overall, this visualization confirms the presence of clear market tiers and technological stratification within the EV dataset.

7. Conclusion

The Electric Vehicle Population dataset implies clear relationship between pricing, technology, and market segmentation. Base MSRP values range widely, and there are clean clusters of Electric Range between short-range PHEVs and long-range BEVs. Weak correlations among numerical variables indicate that EV performance cannot be explained by a single factor alone.
Regression analysis confirmed that Model Year, Base MSRP, and manufacturer jointly explain 77.4% of range variability, motivating technological advancement and brand innovation. Clustering analysis (k = 5) segmented the EV market into identifiable tiers such as affordable hybrids to advanced long-range BEVs. This analysis provides a data-driven understanding of the EV market, demonstrating how technological evolution, pricing strategy, and manufacturer innovation collectively shape EV car market trends.

References

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Appendices

Appendix A

List of Tables

This appendix presents a comprehensive list of all tables included in the *Electric Vehicle Population Data Analysis* report. Each table provides quantitative insights derived from statistical and machine learning analyses, including correlation assessment, regression modelling, and clustering. The listed tables summarize key numerical relationships and outcomes supporting the findings discussed in the main body of the report.

Appendix B

List of Figures

This appendix provides a complete list of all figures included in the *Electric Vehicle Population Data Analysis* report. The figures visually represent key findings from the dataset, highlighting patterns, relationships, and distributions related to electric vehicle characteristics such as model year, manufacturer, base MSRP, and electric range. Each figure supports the analytical discussion presented in the main sections of the report by illustrating data trends and clustering patterns derived from statistical and machine learning methods.

Appendix C

Google Colab Code Repository

This appendix provides access to the Python code developed and executed in Google Colab for the *'Electric Vehicle Population Data Analysis'* project. The Colab notebook contains all scripts, visualizations, and analytical workflows referenced throughout the report. The code supports data cleaning, exploratory analysis, regression modelling, and clustering. It also includes annotated markdown cells explaining each step, ensuring full reproducibility of the analytical process.

C1. Repository Access

The complete Colab notebook can be accessed at the following link:
Google Colab Code: https://tinyurl.com/4prmmz37

C2. Notebook Overview

1. Data Import and Cleaning - Loading the EV population dataset, handling missing values, and preparing features for analysis.
2. Exploratory Data Analysis (EDA) - Visualizing distributions, relationships, and correlations using matplotlib and seaborn.
3. Regression Analysis - Applying linear regression to model Electric Range based on Model Year, Base MSRP, and Make.
4. Clustering Analysis - Using K-Means clustering to segment vehicles by price and range characteristics.
5. Visualization Outputs - Generating comparative plots and summary charts for interpretive insights.

C3. Purpose and Reproducibility

The Colab notebook serves as a transparent and reproducible record of the analytical procedures. It demonstrates technical implementation using Python 3.11 and core data science libraries (pandas, numpy, matplotlib, seaborn, and scikit-learn).